



Scalability & Future Plan

Date	2 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well Being
Maximum Marks	1 Mark

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Predictions are limited to one country at a time	Users cannot analyze multiple countries in bulk within a single request	Medium
2	No user authentication or personalized dashboards	All users share the same view with no saved preferences or history	Low
3	Model is retrained manually and infrequently	Predictions may not reflect the most recent well-being indicator data	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Flask development server handling limited concurrent requests	Deploy behind a load balancer with multiple Gunicorn workers / containerized instances to handle higher concurrent traffic
2	Data Storage	Well-being indicator data stored in a flat CSV file and a local SQLite database	Migrate to a managed cloud database (e.g., PostgreSQL) to support larger datasets and concurrent access
3	Performance	Model loaded and inference executed per request on a single thread	Cache the trained model in memory at startup, use asynchronous request handling, and add response caching for repeated queries
4	Security	No authentication; application publicly accessible via a temporary ngrok tunnel	Add HTTPS, user authentication, rate limiting, and stronger input validation before wider public rollout

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Automated model retraining pipeline using the latest well-being indicator data	Q3 2026	Keeps predictions accurate and up to date
Phase 2	Multi-country bulk upload and batch prediction	Q4 2026	Enables policymakers to analyze many countries at once

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 3	User authentication and saved/personalized dashboards	Q1 2027	Secure, personalized access for different stakeholder roles
Phase 4	Cloud deployment with auto-scaling and public API access	Q2 2027	Supports larger user bases and integration with external research tools